Quantum preprocessing for Internet of Things edge computing security devices

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***Abstract*—Quantum preprocessing has the potential for significantly reducing computing power and storage space needed for tiny devices, such as Internet of Things devices, to satisfactorily operate as security devices such as an Intrusion Detection System. Specifically, we focus on Intrusion Detection Systems (IDS), which analyze network traffic to detect suspicious activities and potential security breaches. Traditional IDS implementations require substantial resources, making them difficult to deploy on edge devices. This paper proposes a quantum process to select critical features from the NSL-KDD dataset for IDS implementation. Our method leverages Grover's algorithm combined with the Pearson correlation coefficient to identify the most relevant features for threat detection. The quantum algorithm executes feature selection through oracle creation and Grover diffusion processes, significantly reducing the dimensionality of the input data while preserving important security indicators. The selected features are evaluated based on quantum measurement results and domain specific security knowledge, considering quantum inaccuracy and hardware resource requirements. Our approach enables IoT devices to operate as effective IDS systems by avoiding the need to process the entire dataset during training, thereby optimizing efficiency and resource usage, further enabling the capability of resource constrained edge devices to operate robust IDS systems.**

Keywords—quantum computing, Internet of Things, cybersecurity, intrusion detection systems, Grover's algorithm, feature selection

I. Introduction

As quantum computing continues to mature, a significant challenge is in the application of the unique strengths of quantum processing onto real-world problems. This problem is so important that there has been significant work in the literature to attempt to uncover salient avenues for quantum exploitation, including Machine Learning (ML) [1] and Artificial Intelligence (AI) [2], energy storage [3], and more. Google, in cooperation with GESDA and XPRIZE in March 2024 announced a $5 million prize for quantum algorithms that can help solve real-world problems [4].

One avenue for exploiting the benefits of quantum computing is in using a quantum computer to do some amount of preprocessing work whose results are then fed into classical computers so that the benefits of each computing paradigm can work together to solve larger problems that they might not otherwise be able to on their own.

In this paper, we offer the idea of using quantum preprocessing in order to enable Internet of Things (IoT) and other embedded systems enhanced capabilities to operate as security devices such as an Intrusion Detection System (IDS). Doing so allows us to open up edge computing security applications to a wider variety of devices and allows us to develop new possibilities for what IoT network security could look like: low power, with minimal or no loss of fidelity.

We investigate whether Grover's algorithm can provide enhanced threat detection within an IDS framework suitable for running on IoT and other embedded devices. Our research objectives focus on three main goals: (1) evaluating the practicality of integrating Grover's algorithm in an IDS context, (2) assessing the speedup and detection performance gained from quantum-enhanced techniques, and (3) establishing the significance of quantum feature selection in improving classification accuracy and resource efficiency. By highlighting these aspects, we aim to clarify the impact of quantum methods in addressing the existing challenges in intrusion detection and provide insights into how quantum computing can reshape IoT edge computing security devices in the near future.

Our contribution is in demonstrating the potential for this kind of quantum preprocessing for IoT edge computing security devices. We walk through how our IBM Quantum System One machine was able to reduce the number of attributes needed to make decision on traffic data from the Network Security Laboratory - Knowledge Discovery and Data Mining (NSL-KDD) dataset [5], allowing for viable decision-making data to be placed on IoT devices with limited storage resources. We will discuss the level of fidelity provided by this quantum preprocessing. We will end with providing suggestions towards replicating the process for other types of security devices that traditionally require large volumes of data and AI/ML processing.

It is our hope that this paper spurs discussion towards new directions for the potential of quantum preprocessing for IoT edge computing security devices.

II. Background

The rapid escalation of cyber threats in recent years has driven the cybersecurity community to seek innovative solutions that can detect and mitigate hostile activities in increasingly complex network environments. Traditional IDS are effective to a certain extent, but they frequently encounter limitations in

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processing massive volumes of data quickly and accurately [6]. Against this, the appearance of quantum computing offers considerable promise to transforming security mechanisms, especially in tasks where classical computing methods struggle with computational restrictions.

A. Classical Feature Selection in IDS

An IDS often operates on large datasets with numerous features, ranging from packet header attributes to derived traffic metrics and behavioral indicators. Classical feature selection aims to reduce the capacity of such datasets by identifying the most important features that contribute to accurate threat detection. Common methods include filter techniques (*e.g.*, mutual information, correlation-based selection), wrapper methods (*e.g.*, genetic algorithms, forward/backward selection), and embedded methods (*e.g.*, decision tree- or regularization-based selection). These approaches can substantially lower computational overhead by filtering out irrelevant features, therefore improving the efficiency of machine learning models and minimizing false positives. Nevertheless, they can be computationally expensive, especially in large-scale network environments where iterative selection processes must be repeated frequently to adapt to evolving threats [7].

B. Quantum Computing Fundamentals and Challenges

Quantum computing harnesses the principles of quantum mechanics—superposition (the ability of quantum bits to exit in multiple states simultaneously), entanglement (quantum correlations between particles), and interference (the interaction between quantum states) —to address problems more efficiently than classical systems. Within cybersecurity, quantum algorithms have shown promising advances in areas such as cryptography, optimization, and pattern recognition, thereby opening avenues for more robust defenses against evolving cyber threats. Quantum algorithms like Grover's can achieve quadratic speedups when searching through unstructured datasets of potential malicious behaviors [8]. By leveraging Grover's algorithm, cybersecurity applications can accelerate certain analysis and detection processes that benefit from searching through large datasets of potential malicious behaviors.

Despite the potential of quantum approaches, several challenges remain in implementing network intrusion detection at scale. Intrusions today often occur in distributed, high efficiency environments, demanding both rapid response and adaptability. This not only stresses computational resources but also underscores the need for feature selection techniques that can identify the most relevant indicators of malicious activity. Quantum feature selection—taking advantage of quantum-enhanced searching and optimization—could streamline the detection process, filtering out extraneous data and focusing on critical threat signatures more effectively than classical approaches [9, 10].

While quantum algorithms offer theoretical advantages, their practical implementation faces several hardware challenges. Modern quantum processors, often referred to as Noisy Intermediate-Scale Quantum (NISQ) devices, support only a limited number of qubits [11]. These qubits are prone to decoherence, a phenomenon where quantum states gradually lose their coherence due to interactions with the environment. Additionally, gate operations have non-zero error rates, and error-correction schemes remain resource-intensive, demanding more qubits than are typically available. Furthermore, environmental noise and qubit cross-talk can degrade algorithmic performance, complicating direct application of quantum algorithms in production-scale settings. As a result, research in this area often involves hybrid approaches, employing classical post-processing techniques to mitigate hardware-specific limitations while leveraging quantum speedups.

C. Quantum Machine Learning Approaches

Despite these hardware limitations, researchers have developed several promising quantum machine learning (QML) approaches that work within current technological constraints, such as anomaly detection, malware classification, and cryptanalysis. Quantum Support Vector Machines (QSVM) [12], Quantum Neural Networks (QNN) [13], and Quantum Boltzmann Machines (QBM) [14] are among the frameworks under investigation for their potential to achieve faster training or more nuanced pattern recognition than their classical counterparts. Early studies suggest that quantum algorithms may excel in high-capacity data scenarios, offering a path toward more robust detection of subtle intrusions and zero-day attacks. However, most existing research remains in the proof-of-concept stage, constrained by the factors mentioned above—particularly limited qubit counts and noise-sensitive quantum operations. Ongoing efforts seek to bridge this gap by developing noise-resilient algorithms and near-term quantum-classical hybrid strategies to fully harness the potential of quantum computing in cybersecurity. Additionally, little attention has been paid to enhancing IoT and other embedded devices, something we hope to be able to address.

III. METHODOLOGY

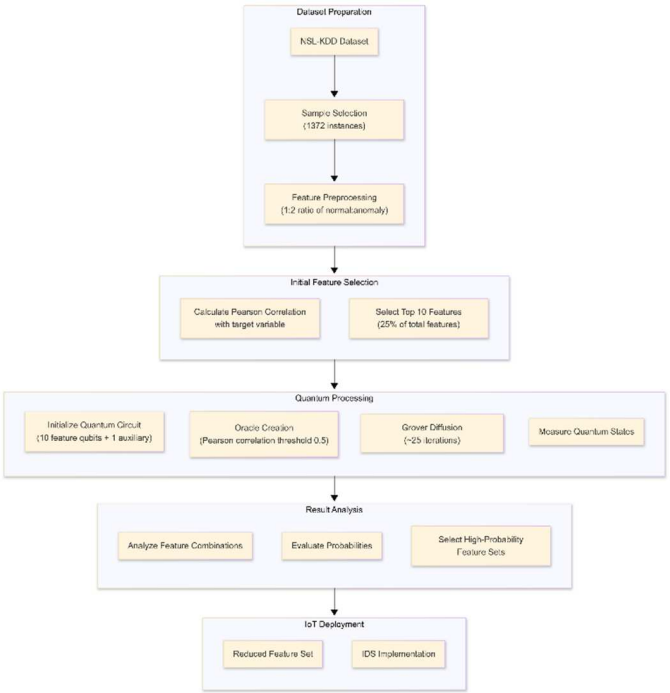


Figure 1. Block diagram illustrating the overall methodology

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A. *Dataset preparation and preprocessing*

The NSL-KDD dataset was used for feature selection. Derived from the KDDCUP '99 dataset [15], a benchmark for anomaly detection since 1999, NSL-KDD addresses its predecessor's limitation of duplicate records and an unbalanced sample ratio, both of which affect the accuracy of the learning algorithm [16]. Tavallaee et al. improved the dataset by removing redundant records and rebalancing the data distribution [5]. Specifically, they selected records inversely proportional to their difficulty levels—including more records from underrepresented difficult cases and fewer from overrepresented easy cases in the original KDDCUP '99 dataset [17]. We chose NSL-KDD for its status as a standard IDS evaluation benchmark [18] and its accessibility.

Due to the hardware limitations of the quantum computer in terms of computation power and the innate existence of quantum noise, a reasonable number of samples is set to be sampled for each run of the algorithm. To deduce the number of samples that would be sufficient for providing a statistically significant result, past literature was consulted. In another study that attempted to find significant features for IDS using the KDD dataset using statistical methods [19], different sample sizes were used across all the experimentations of different selected features. Comparing the percentages of correctly classified samples for systems trained with different sample sizes, the small samples consisting of 1372 instances performed almost on par with larger sample sizes, achieving 93-94% accuracy, only around a few percentages lower than higher sample sizes. Hence, this sample size was used for this paper's experimentation. That we can get small sample sizes to perform so close to larger models suggests that smaller IoT and embedded devices should be able to perform detection adequately after given the outputs from the quantum preprocessing.

The selection process is completed through computation. The percentage of each feature with the sample size mirrors that of the original KDD dataset, while the selection of samples within each feature is randomized for each experimentation run, a selection of both normal and anomalies in a 1 to 2 ratio is retained.

B. *Feature Selection Criteria and Quantum Process*

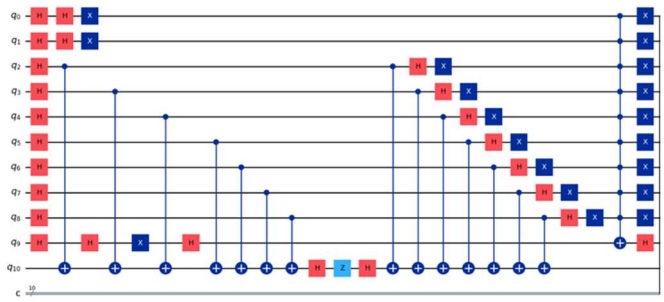


Figure 2: Quantum circuit for Grover-based feature selection implemented in Qiskit. The circuit uses q0-q9 as feature qubits and q10 as auxiliary qubit. Red squares represent Hadamard gates, dark blue squares represent NOT gates, light blue squares represent phase gates, and vertical lines with filled/cross ends represent CNOT gates.

The feature selection uses Grover's algorithm with Pearson correlation as the statistical metric. Each feature qubit can be $|0\rangle$ or $|1\rangle$, with $|1\rangle$ denoting the selection of such feature. With n features, there are 2^n possible states, each with a unique combination of features. For example, for three features, and a state of $|100\rangle$, the first feature is selected while then second and second and third are not, hence only the first qubit is $|1\rangle$. The feature selection process begins by creating a superposition of all feature combinations.

B1. *Oracle creation*

The absolute Pearson correlation coefficient is calculated between each feature and the target variable (normal/anomaly label for each sample). This correlation measures how strongly features and target values move together—whether higher feature values tend to occur with anomalies and lower values with normal samples, or vice versa.

Features with correlation above a certain threshold are "marked" by connecting their qubit to the auxiliary qubit using CNOT gates. These gates flip the auxiliary qubit when the feature qubit is $|1\rangle$, providing the necessary quantum reversibility. A phase kickback sequence (H-Z-H gates) is then applied, attaching a -1 phase factor to states with flipped auxiliary qubits and +1 to others. States with the -1 phase will have their amplitudes increased during amplitude amplification, making the quantum system more likely to measure states corresponding to good feature combinations. The oracle concludes by uncomputing the CNOT gates to restore the auxiliary qubit.

From Figure 2, it can be seen that correlation-based marking is implemented through a series of CNOT gates which connects the auxiliary qubit with a target qubit based on their correlation strength with the target variable. The phase kickback sequence is seen to be applied on the auxiliary qubit using a series of H-Z-H gates.

B2. *Grover diffusion*

Following the oracle creation, the Grover diffusion operator is then applied to all the feature qubits. The operator utilizes the meaningful phase factors and amplifies specific states while decreasing others. First, all qubits are converted to Hadamard basis, transforming computational basis states into superposition states, making it easier to perform the reflection operation. X gates then flip all qubits, preparing the state for phase inversion around $|0\rangle$. A H-MCX-H sequence follows, using the last feature qubit as the target and all other feature qubits as controls. This implements a controlled-Z operation that creates the phase inversion around $|0\rangle$ which, combined with the Hadamard transformations, creates the reflection about the uniform superposition. The X gates are then uncomputed, returning qubits to their pre-flip states. Finally, the qubits are transformed from Hadamard basis back to computational basis, completing the diffusion operation.

After the initial superposition states, the oracle and diffusion are repeated based on a set number of iterations.

From Figure 2, the Grover diffusion operator can be observed as the H-X operations on all qubits, followed by a central multi-controlled operation (H-MCX-H sequence) where multiple feature qubits on different qubit lines connect vertically to the target qubit. Uncomputation of the X gates and the final Hadamard transform completes one iteration.

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IV. Experimental Setup

The quantum computer used for experimentation is the IBM Quantum System One installed at Rensselaer Polytechnic Institute in April 2024. It has a total of 127 qubits with 32,000 CLOPS. In addition, all the code for data loading, preparation and function writing that heavily uses the IBM Qiskit library running on a laptop.

The optimal number of iterations were calculated using the formula:

$$\frac{\pi}{4}\sqrt{N}$$

Where N is the total number of possible states. Given that the that the total number of states is 2^n, where n is the number of features, fully utilizing all the features in the NSL-KDD dataset where n=41 will result in:

$$N_{optimal} = \frac{\pi}{4}\sqrt{2^{41}} \approx 1{,}164{,}675$$

This number of iterations is too high to realistically implement on a quantum computer, hence the Pearson correlation was calculated to pre-select the top 10 features, about 25% of the total number of features. With 10 features, the optimal number of iterations will lower to:

$$N_{optimal} = \frac{\pi}{4}\sqrt{2^{10}} \approx 25$$

Although limiting the number of features can cause a degree of inaccuracy, it is generally still better than lowering the number of iterations, which could deviate from achieving sufficient amplitude amplification in the Grover diffusion and an optimal Grover search.

With this design, only 11 qubits out of the total 127 to represent the 10 features + 1 auxiliary qubit are needed. This number balances the hardware limitation of higher error rates at higher qubits and algorithmic requirements with the number of features, and the computational feasibility in terms of the number of iterations.

A. Correlation Threshold for Oracle Creation in Feature Selection

The correlation threshold is set to be 0.5 for marking good features for several reasons. Statistically, a 0.5 Pearson correlation indicates a moderate to strong relationship between the feature and target (normal/anomaly) variables [20], where approximately 25% of the variance ($r^2 = 0.25$) in the target variable can be explained by the feature. According to Cohen's correlation guidelines, a 0.5 correlation threshold is considered as the beginning of larger effects [21]. In practice, 0.5 should provide a good balance between filtering out weak/noisy and potentially useful features.

B. Performance metrics

The performance metrics would be runtime, error rates, and classification accuracy. We want to explore the benefits of using a quantum computer for preprocessing rather than using a classical computer, recording the total time for running the algorithm would be a key statistic. Quantum hardware errors are inherent within the operations in each quantum gate, while other errors such as decoherence, measurement and connection errors also exist. Meanwhile, classification errors would arise upon using the selected features in an IDS system, depending on whether the system can detect anomalies accurately. However, since this paper focuses more on the feature selection algorithm rather than the application of its product, the good features found by the quantum computer will only be scrutinized under a technical and domain specific sense, on whether the algorithm was able to find distinct, meaningful combinations that can be useful for IDS.

V. Results and Analysis

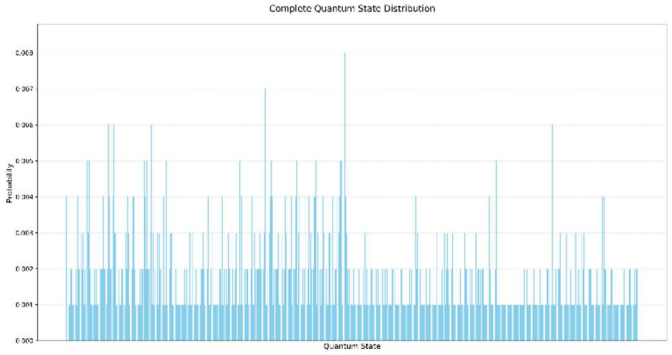


Figure 3: Quantum state probability distribution for all the possible states. Each bar represents a feature combination. States with higher probability suggest significant feature combinations. The x-axis labels for each state is not shown due to the density of the bars.

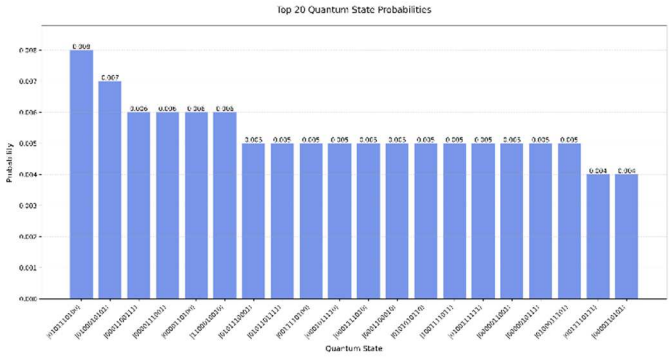


Figure 4: A subset of the total quantum state probability distribution with only the top 20 states with the highest probabilities.

The combination of features with the highest probabilities are:

0.008: is_guest_login + is_host_login + rerror_rate + srv_serror_rate + su_attempted

0.007: dst_host_serror_rate + is_guest_login + is_host_login + su_attempted

From a technical sense, the algorithm successfully identified meaningful feature combinations. There were many noticeable peaks in the quantum state probability distribution, meaning that the quantum circuit effectively amplified important feature patterns. The algorithm also found combinations of varying lengths for the highest probabilities, suggesting that the quantum search space was overall well-explored across different dimensionalities. Meanwhile, the combinations show both direct

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attack indicators like login attempts, and supporting indicators like error rates and command patterns, while avoiding redundancies in combinations. This suggests effective capture of feature interactions.

From a domain specific perspective, the first combination integrates authentication patterns (is_guest_login, is_host_login) with error rates (rerror_rate, srv_serror_rate) and privilege escalation attempts (su_attempted). This creates a comprehensive detection pattern for complex Root to Local (R2L) attacks where attackers attempt password guessing, generate connection errors during probing, and try to escalate privileges after gaining initial access [22]. The quantum algorithm assigned this a higher probability because it captures the complete attack chain from initial access to privilege escalation. The second combination focuses on host-based attack patterns by combining host-level error rates (dst_host_serror_rate) with authentication features and privilege escalation attempts. This combination is particularly effective at detecting distributed attacks where adversaries attempt to compromise multiple hosts through guest/host login mechanisms before attempting privilege escalation. Its slightly lower probability reflects its more specific focus on host-based attack scenarios versus the broader attack coverage of the first combination. Both combinations demonstrate the algorithm's ability to identify feature interactions that map to real-world attack patterns, particularly the progression from initial access to privilege escalation attempts.

In terms of runtime, the entire circuit ran for 21 seconds on the IBM Quantum System One, while the entire code ran in under 3 minutes including the time waiting for execution on quantum hardware. Such efficiency can mitigate worries about resource usage on both the classical and quantum computer.

Lastly, in terms of error rates, the quantum computer was able to generate a few distinct combinations with notably higher probabilities (0.008 and 0.007) as seen from Figure 3, but from Figure 4 it can be seen that there remains a considerable population of states with probabilities just slightly lower than these peaks. This distribution pattern suggests that while the quantum circuit successfully amplified certain feature combinations, quantum noise and gate errors may be affecting the sharpness of this amplification [23]. Nevertheless, the error profile is consistent with running our algorithm on NISQ-era hardware, where perfect isolation of optimal states is challenging to achieve [24].

VI. Discussion

The main advantage of Grover's algorithm is its theoretical quadratic speedup in unstructured search. In the context of feature selection, this could mean more efficient scans through high-dimensional feature spaces compared to classical approaches. The benefits in current applied experiments remain bounded by two major factors: dataset size and qubit availability.

From a dataset perspective, the number of features n grows linearly with the qubits needed for the system (n feature qubits plus at least one auxiliary qubit). While current quantum hardware nominally provides up to 127 qubits, applied use cases are often limited to utilizing only around 6 qubits due to high error rates. While Grover's algorithm scales favorably in theory, the practical realization will need further hardware and algorithmic optimizations to handle the broader feature sets typical of enterprise-level network traffic data.

A. Hardware limitations and mitigation

Current quantum hardware creates constraints in terms of qubit count, qubit connectivity, gate fidelity, and decoherence times. The steps in Grover's algorithm, such as multiple iterations of oracle marking and diffusion, rapidly accumulate gate operations, increasing susceptibility to noise and errors.

Circuit optimization and transpilation can help decrease the effects by reducing the total number of quantum gates through improved circuit layouts. Moreover, noise-aware qubit mapping, which involves selecting qubits with higher fidelities and employing sophisticated routing schemes, can minimize inter-qubit errors and overall gate overhead.

In addition to these hardware-sensitive strategies, it is essential to manage errors effectively. Qiskit documentation distinguishes error mitigation, which aims to correct or reduce errors post-measurement [25], from error suppression, which focuses on minimizing errors before or during circuit execution [26]. Since the Qiskit sampler currently does not support error mitigation, we used various error suppression techniques such as dynamical decoupling and Pauli twirling to reduce coherent noise and gate infidelity at the hardware level. While these methods do not reduce the circuit's measurement overhead in the same way full mitigation might, they can significantly improve overall fidelity, especially under near-term hardware constraints.

B. Hybrid Approach and Quantum Advantages

While fully quantum-based intrusion detection systems may still be a long way off, hybrid quantum-classical approaches could become viable much sooner. By allowing quantum circuits to identify top candidate features and then relying on classical methods for refinement and validation, these hybrid strategies can markedly reduce the search space for critical threat indicators. Moreover, faster re-training cycles enabled by this division of labor are key for keeping pace with an ever-changing threat environment and improved detection accuracy. Fewer false positives can be achieved without the full overhead of a purely quantum solution. As quantum hardware progresses, the synergy between quantum-enhanced feature selection and classical processing promises to become an increasingly integral part of next-generation IDS frameworks.

Quantum preprocessing also offers several advantages compared to purely classical methods. First, the quantum feature selection cost occurs only once—results can then be distributed to many IoT devices. Running the algorithm also incurs little to no time, as evidenced by the 21 second runtime. Second, devices only need to store selected features rather than the full dataset, significantly reducing resource requirements. Devices only need to implement simpler detection logic focused on key indicators. Third, quantum superpositions enable identification of non-obvious feature correlations that might be missed by classical methods. Finally, the approach offers scalability benefits as quantum hardware continues to improve.

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C. Future research directions

One direction that is currently being looked at is improving oracle construction. Adaptive thresholding, where thresholds are selected based on statistical significance tests or dynamically inferred from the data, may offer more accurate detection of correlated features than a fixed cutoff. There are also algorithmic variants that extend Grover's approach to partial diffusion or multi-solution oracles [27], which may provide better feature selection by accommodating multiple relevant feature subsets.

Further qubit optimization and circuit depth reduction could enable the algorithm to scale beyond small experiments. Techniques such as qubit reuse [28], gate decomposition optimizations [29], or specialized layout adaptations help minimize error accumulation and qubit resource usage.

Finally, comprehensive performance evaluation studies are needed to validate the effectiveness of the selected features. Future experiments should systematically compare our quantum preprocessing approach against established classical features selection techniques, including filter methods (mutual information [30], chi-squared testing [31]), wrapper methods (forward/backward selection [32]), and embedded methods (LASSO [33], decision tree-based selection [34]). These comparisons should evaluate computational overhead, feature selection time, and feature set quality. The experiments should assess IDS performance using quantum-selected feature combinations against the full NSL-KDD feature set and classically selected features, examining detection accuracy, false positive rates and resource utilization in IoT contexts. Real-world validation should also be carried out on actual IoT devices to measure detection capabilities, resource consumption and processing latency, while generalization capabilities should be assessed using datasets beyond the NSL-KDD.

Such studies would clarify the true advantages of quantum-assisted pipelines and guide practitioners in determining if, when, and how to deploy quantum approaches in a production-grade environment, particularly in resource constrained IoT environments.

VII. Conclusion

This article contributes to the literature in applied quantum security research by demonstrating the potential for quantum preprocessing to be a viable avenue for research and practicality to enhance IDS framework development and leveraging outputs to enable IoT edge computing security in new and novel ways. In addition to furthering this potential in future work, we also seek to create simplified versions of this avenue of research suitable for teaching undergraduate and younger students applied quantum security. Indeed, this type of work is already rapidly being done broadly in the quantum space [35] and the work described in this paper can be simplified to be added to this corpus of research.

References

[1] A. Perdomo-Ortiz, M. Benedetti, J. Realpe-Gómez, and R. Biswas, "Opportunities and challenges for quantum-assisted machine learning in near-term quantum computers," Quantum Science and Technology, vol. 3, June 2018. doi: 10.1088/2058-9565/aab859.

[2] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Peterson, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," Nature vol. 518, pp. 529-533, February 2015. doi: 10.1038/nature21056.

[3] A. Ho, J. McClean, and S. P. Ong, "The promise and challenges of quantum computing for energy storage," Future Energy, vol. 2, pp. 810-813, May 2018. doi: 10.1016/j.joule.2018.04.021.

[4] B. H. Gosselink and R. Babbush, "Google, GESDA, and XPRIZE launch new competition in quantum applications," Google, March 2024. [Online] Available: https://blog.google/technology/research/google-gesda-and-xprize-launch-new-competition-in-quantum-applications/

[5] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," *2009 IEEE symposium on computational intelligence for security and defense applications*, pp. 1-6, 2009. doi: 10.1109/CISDA.2009.5356528.

[6] A. Fenjan, M. T. M. Almashhadany, S. R. Ahmed, H. A. Fadel, R. Sekhar, P. Shah, and B. S. Veena, "Adaptive intrusion detection system using deep learning for network security," AICCONF '24: Proceedings of the Cognitive Models and Artificial Intelligence Conference, pp. 279-284, May 2024. doi: 10.1145/3660853.3660928.

[7] J. Lansky, S. Ali, M. Mohammadi, M. K. Majeed, S. H. T. Karim, S. Rashidi, M. Hosseinzadeh, and A. M. Rahmani, "Deep learning-based intrusion detection systems: A systematic review," in *IEEE Access*, vol. 9, pp. 101574-101599, 2021. doi: 10.1109/ACCESS.2021.3097247

[8] L. K. Grover, "Fixed-point quantum search," Physical Review Letters, vol. 95, pp. 150501-1–150501-4, October 2005. doi: 10.1103/PhysRevLett.95.150501.

[9] Z. He, L. Li, Z. Huang, and H. Situ, "Quantum-enhanced feature selection with forward selection and backward elimination," Quantum Information Processing, vol. 17, May 2018. doi: 10.1007/s11128-018-1924-8.

[10] S. Mücke, R. Heese, S. Müller, M. Wolter, and N. Piatkowski, "Feature selection on quantum computers," Quantum Machine Intelligence, vol. 5, February 2023, doi: 10.1007/s42484-023-00099-z.

[11] J. Preskill, "Quantum computing in the NISQ era and beyond," Quantum: The open journal for quantum science, vol. 2, July 2018. doi: 10.22331/q-2018-08-06-79.

[12] T. Suzuki, T. Hasebe, and T. Miyazaki, "Quantum support vector machiens for classification and regression on a trapped-ion quantum computer," Quantum Machine Intelligence, vol. 6, May 2024. doi: 10.1007/s42484-024-00165-0.

[13] S. Gupta and R. K. P. Zia, "Quantum neural networks," Journal of Computer and System Sciences, vol. 63, pp. 355-383, November 2001. doi: 10.1006/jcss.2001.1769.

[14] M. H. Amin, E. Andriyash, J. Rolfe, B. Kulchytskyy, and R. Melko, "Quantum Boltzmann Machine," Physical Review X, vol. 8, pp. 021050-1–021050-11, May 2018. doi: 10.1103/PhysRevX.8.021050.

[15] S. Hettich and S. D. Bay, "The UCI KDD Archive," Irvine, CA: University of California, Department of Information and Computer Science, 1999. [Online] Available: https://kdd.ics.uci.edu/

[16] Y. Tang, L. Gu, and L. Wang, "Deep Stacking Network for Intrusion Detection," Sensors (Basel), vol. 22, p. 25, December 2021. doi: 10.3390/s22010025.

[17] L.Dhanabal and D. S. P. Shantharajah, "A Study on NSL-KDD Dataset for Intrusion Detection System Based on Classification Algorithms," 2015. Accessed: Dec. 28, 2024. [Online]. Available: https://www.semanticscholar.org/paper/A-Study-on-NSL-KDD-Dataset-for-Intrusion-Detection-L.Dhanabal-Shantharajah/f00d9a1533b296a878656db4589246a0e1597db0

[18] K. Dinesh and D. Kalaivani, "Enhancing Performance of Intrusion detection System in the NSL-KDD Dataset using Meta-Heuristic and Machine Learning Algorithms-Design thinking approach," 2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS), June 2023, pp. 1471–1479. doi: 10.1109/ICSCSS57650.2023.10169845.

[19] S. Zargari and D. Voorhis, "Feature Selection in the Corrected KDD-dataset," in 2012 Third International Conference on Emerging Intelligent Data and Web Technologies, September 2012, pp. 174–180. doi: 10.1109/EIDWT.2012.10.

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[20] H. Akoglu, “User’s guide to correlation coefficients,” Turk J Emerg Med, vol. 18, no. 3, pp. 91–93, Aug. 2018, doi: 10.1016/j.tjem.2018.08.001.

[21] G. E. Gignac and E. T. Szodorai, “Effect size guidelines for individual differences researchers,” Personality and Individual Differences, vol. 102, pp. 74–78, November 2016, doi: 10.1016/j.paid.2016.06.069.

[22] S. K. Pandey, “Design and performance analysis of various feature selection methods for anomaly-based techniques in intrusion detection system,” SECURITY AND PRIVACY, vol. 2, no. 1, p. e56, 2019, doi: 10.1002/spy2.56.

[23] A. A. Clerk, M. H. Devoret, S. M. Girvin, F. Marquardt, and R. J. Schoelkopf, “Introduction to quantum noise, measurement, and amplification,” Rev. Mod. Phys., vol. 82, no. 2, pp. 1155–1208, Apr. 2010, doi: 10.1103/RevModPhys.82.1155.

[24] D. Koch, B. Martin, S. Patel, L. Wessing, and P. M. Alsing, “Demonstrating NISQ era challenges in algorithm design on IBM’s 20 qubit quantum computer,” AIP Advances, vol. 10, no. 9, p. 095101, Sep. 2020, doi: 10.1063/5.0015526.

[25] “Configure error mitigation | IBM Quantum Documentation,” IBM. [Online] Available: https://docs.quantum.ibm.com/guides/configure-error-mitigation.

[26] “Error mitigation and suppression techniques | IBM Quantum Documentation,” IBM. [Online] Available: https://docs.quantum.ibm.com/guides/error-mitigation-and-suppression-techniques.

[27] D. Konar, Z. Hafeez, and V. Aggarwal, “A bi-directional quantum search algorithm,” arXiv:2404.15616v1 [quant-ph], April 2024. doi: 10.48550/arXiv.2404.15616.

[28] M. DeCross, E. Chertkov, M. Kohagen, and M. Foss-Feig, “Qubit-reuse compilation with mid-circuit measurement and reset,” Physical Review X, vol. 13, pp. 041057-1–041057-22, December 2023. doi: 10.1103/PhysRevX.13.041057.

[29] J. Chi, X. He, Y. Zhou, J. Yuan, L. Zhang, Q. Guo, Y. Hai, Z. Han, C-K. Hu, W. Huang, H. Jia, D. Jiao, S. Li, Y. Liu, Z. Ni, L. Nie, X. Pan, J. Qiu, W. Wei, W. Nuerbolati, Z. Yang, J. Zhang, Z. Zhang, W. Zou, Y. Chen, X. Deng, X. Deng, L. Hu, J. Li, S. Liu, Y. Lu, J. Niu, D. Tan, Y. Xu, T. Yan, Y. Zhong, F. Yan, X. Sun, and D. Yu, “Scalable algorithm simplification using quantum AND logic,” Nature Physics, vol. 19, 126-131, November 2022. doi: 10.1038/s41567-022-01813-7.

[30] M. Beraha, A. M. Metelli, M. Papini, A. Tirinzoni, and M. Restelli, “Feature Selection via Mutual Information: New Theoretical Insights,” Jul. 17, 2019, arXiv: arXiv:1907.07384. doi: 10.48550/arXiv.1907.07384.

[31] I. Sumaiya Thaseen and C. Aswani Kumar, “Intrusion detection model using fusion of chi-square feature selection and multi class SVM,” Journal of King Saud University - Computer and Information Sciences, vol. 29, no. 4, pp. 462–472, Oct. 2017, doi: 10.1016/j.jksuci.2015.12.004.

[32] M. A. Thanoon, M. J. M. Zedan, and A. N. Hameed, “Feature Selection Based on Wrapper and Information Gain,” in 2019 1st AL-Noor International Conference for Science and Technology (NICST), Oct. 2019, pp. 32–37. doi: 10.1109/NICST49484.2019.9043805.

[33] R. Muthukrishnan and R. Rohini, “LASSO: A feature selection technique in predictive modeling for machine learning,” in 2016 IEEE International Conference on Advances in Computer Applications (ICACA), Oct. 2016, pp. 18–20. doi: 10.1109/ICACA.2016.7887916.

[34] K. Grabczewski and N. Jankowski, “Feature selection with decision tree criterion,” in Fifth International Conference on Hybrid Intelligent Systems (HIS’05), Nov. 2005, p. 6 pp.-. doi: 10.1109/ICHIS.2005.43.

[35] B. R. Callahan, K. Schilp, Q. Colognato, E. Goldman, S. Sugerman, A. Mehta, A. Imanuel, K. Kaii, and H. Rose, “Multidisciplinary quantum cybersecurity research for the undergraduate laboratory,” The Journal of the Colloquium for Information Systems Security Education, *forthcoming*.

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